
InfoServer

**A GPS-Disciplined Stratum-1 Network Time and
Information Server**

Operator Manual

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This manual covers the setup, configuration, and day-to-day operation of the InfoServer — a self-contained network time server that derives time from a GPS receiver (or an upstream Internet time server), disciplines a temperature-compensated DS3231 real-time clock to that source, and serves it to NTP clients on your local network. It also publishes local weather, sensor readings, news headlines, and nearby aircraft on your network and can alert you by time-source faults. It includes a Theory of Operation chapter explaining how the server produces and distributes accurate time.

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Quick Start

The InfoServer is a stratum-1 network time server. It keeps highly accurate time from a GPS receiver (or, as a fallback, from an Internet time server) and hands that time out to every computer, clock, and device on your network using the standard NTP protocol. It can also publish local weather and environmental readings on your network, and send you an email or Telegram alert if a time-source problem occurs. A full explanation of every option is in the numbered sections that follow.

1. Position the server and power it

The GPS antenna is built into the receiver, so there is nothing to attach — just place the server itself where it can see the sky, such as a window sill or, ideally, outdoors. Connect a 5 V USB-C power supply. On a brand-new unit, AP setup mode starts automatically (see step 2).

2. Connect to the server's WiFi

From a phone or computer, open WiFi settings and look for the network named after the server — InfoServer by default. Connect to it; no password is required for the setup network.

3. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- Select your home WiFi network from the SSID dropdown and enter its password.
- Leave the Discipline Mode at NMEA GPS unless you intend to run from an Internet time server instead of GPS.

Tap Save, then Start. The server saves the settings and reboots onto your network.

4. Wait for a GPS fix

After reboot the server joins your WiFi and begins serving as soon as a time source is locked. A first GPS fix from a cold start can take from under a minute to several minutes depending on how much open sky the unit can see. The STATUS LED shows progress (Section 9), and the status web page shows the live state.

5. Point your devices at it

On each device, set the NTP server address to this unit's IP address (shown on the status page) or its mDNS name — `infoserver.local` by default, or `<name>.local` if you change the Access Point Name. The server answers standard NTP requests on UDP port 123 — no client software is required beyond the NTP client already built into every operating system.

6. Daily use

To do this	Do this
See live status	Browse to the server's IP address (or <code>http://infoserver.local/</code>) from any device on the same WiFi
Read weather / sensor / news / aircraft data	Fetch <code>http://infoserver.local/info</code> from any device (see Section 7)
Get email or Telegram alerts on time faults	Enable and configure them on the Notifications page (see Section 8)
Show the IP address on the LEDs	Double-click SW1 — the IP blinks out octet by octet
Install new software over WiFi	Hold SW1 for 10 seconds
Re-enter setup (AP) mode	Hold SW1 for 5 seconds, or use Switch to AP Mode on the status page (System sub-page)
Factory reset	Hold SW1 while applying power; the unit wipes its settings and reboots into AP mode

1. Overview

The InfoServer is a self-contained network time and information server built on an ESP32-S3. It derives time from one of two sources — a u-blox GPS receiver or an upstream Internet NTP server — disciplines a temperature-compensated DS3231 real-time clock to that source, and serves the result to NTP clients on your local network. Alongside the time service it can publish environmental and weather information on your LAN as simple JSON, and alert you by email or Telegram message when a time-source problem arises. A web interface handles all configuration; there is no app to install and no cloud account.

Why run your own time server?

- Accuracy. A GPS-disciplined server delivers time within roughly a millisecond of true UTC on your LAN — far tighter than typical Internet sync, and with no dependence on your Internet connection once GPS is locked.
- Independence. Your devices sync to a source on your own network. They stay in agreement with each other even if the Internet is down.
- Privacy and control. NTP requests stay on your LAN rather than going out to public time servers.
- More than time. The same box can serve local weather and sensor readings to your devices and dashboards, and alert you by email or Telegram if timekeeping is disrupted.

What's in the box

- The InfoServer unit (ESP32-S3 controller, DS3231 RTC with battery backup, an optional BME280 environmental sensor, indicator and under-light LEDs).
- A u-blox GPS receiver with a built-in antenna and a 1PPS (one-pulse-per-second) output.
- A USB-C port for power — use any standard 5 V USB-C supply (power supply not included).

Key features

- Stratum-1 GPS discipline using NMEA time plus the 1PPS edge for sub-second accuracy.
- Automatic NTP fallback if GPS is unavailable, with a configurable holdover window.

- Battery-backed DS3231 RTC carries time through brief power and signal interruptions.
- Local information services: a JSON /info manifest publishing on-board sensor readings with a barometric trend, current weather and forecast from up to three providers (OpenMeteo needs no API key), active National Weather Service alerts, locally computed sunrise/sunset and moon-phase data, news headlines from up to five RSS feeds, and aircraft currently flying near your location (Section 7).
- Email and Telegram notifications on time-source faults, with built-in test buttons (Section 8).
- Web-based setup and a live status page; mDNS discovery as infoserver.local by default (follows the Access Point Name).
- Over-the-air software updates.
- Two status indicators plus two color-configurable under-lights.

2. Theory of Operation

This section explains how the InfoServer produces and distributes accurate time. It is background reading — none of it is required to use the server, but it explains what the status page is telling you and why the design behaves as it does.

The problem NTP solves

Every computer has a clock, but those clocks drift — a few seconds a day is typical. The Network Time Protocol (NTP) lets a device ask a time server "what time is it?", measure how long the question-and-answer round trip took, and correct its own clock accordingly. The key trick is that NTP timestamps the request on the way out and the reply on the way back, so the client can subtract the network delay and recover the true time the server held at a known instant. The quality of everyone's clock therefore depends on the quality of the server's clock — which is what this device is built to maximize.

Stratum: the time hierarchy

NTP arranges time sources in numbered tiers called strata. Stratum 0 is a reference clock itself — an atomic clock or, here, the GPS system, whose satellites carry atomic clocks. A server directly attached to a stratum-0 reference is stratum 1. A server that syncs from a stratum-1 server is stratum 2, and so on; each hop adds a little uncertainty. When this unit is GPS-disciplined it advertises stratum 1. When it falls back to an Internet NTP server (typically stratum 2) it advertises stratum 3.

The time source: a disciplined RTC

At the heart of the server is a DS3231 — a temperature-compensated real-time clock chip accurate to roughly two parts per million (about a tenth of a second per day on its own). The DS3231 is not trusted to keep perfect time by itself; instead it is continuously corrected — "disciplined" — by a more accurate source. Between corrections, the server does not read the chip's ticking registers for the served time. It keeps the corrected moment in memory and extrapolates forward using the processor's own millisecond timer, which yields a smooth, phase-accurate clock that can report not just the second but the precise fraction of a second.

Discipline source 1: GPS

In the default mode the discipline source is the GPS receiver. GPS is two signals working together. First, the receiver sends NMEA sentences over a serial line —

ordinary text messages that include the current date and time to the whole second. Second, and crucially, it emits a 1PPS pulse: a sharp electrical edge that occurs exactly on each UTC second boundary, accurate to tens of nanoseconds. The NMEA text says which second it is; the 1PPS edge says exactly when that second begins. The firmware combines the two — taking the whole second from the text and the precise instant from the pulse — to anchor the served clock to true UTC within well under a millisecond. This is what makes the server stratum 1.

Discipline source 2: upstream NTP

If you are running without GPS, or GPS never acquires a fix, the server can instead discipline itself from an Internet NTP server (for example `pool.ntp.org` or `time.google.com`). This is less accurate — Internet round trips introduce some tens of milliseconds of uncertainty — but it needs no view of the sky and still produces a perfectly usable LAN time server. In this mode the unit advertises stratum 3.

Holdover: riding through a GPS dropout

GPS reception is not perfectly continuous — a passing obstruction or a poor view of the sky can briefly interrupt the fix. Because the DS3231 drifts only about seven milliseconds per hour, the server does not panic the instant GPS disappears. It enters holdover: it keeps serving GPS-derived time from the DS3231, still as stratum 1, for a configurable window (one hour to one day, default four hours). Only if GPS stays gone past that window does the server hand discipline to the NTP fallback. The logic deliberately prefers a few hours of holdover over an immediate switch, because the drifting-but-recent GPS time is actually tighter than a fresh Internet NTP sample would be.

NTP as a witness

When both GPS and an upstream NTP server are available, the NTP fallback does not sit idle. While GPS holds discipline, the server still polls the upstream NTP server on a schedule — but only to compare, not to correct. The difference between the two independent sources is shown on the status page as the GPS–NTP figure. In normal operation it is a few milliseconds, which is reassuring evidence that two unrelated time sources agree. If they ever disagree by more than a second and keep disagreeing, the figure turns red — a genuine alarm that something (a GPS fault, poor reception, a misbehaving upstream server) deserves attention.

Guarding against bad data

A time server that blindly accepted every sample could be thrown wildly off by a single garbled GPS sentence or a malformed packet. Two safeguards prevent this. First, GPS sentences are checksum-verified before use, and NTP responses are validated against the protocol. Second, a sanity filter rejects any correction that would jump the clock by more than a minute — unless the same large offset persists across several samples, in

which case it is eventually accepted as a real (if dramatic) correction. The result is a clock that tracks small, legitimate adjustments instantly but refuses to be yanked around by a one-off glitch.

Serving clients

The server listens for NTP requests on UDP port 123. When a request arrives it records the exact moment of arrival, fills in the reply with its reference identifier and stratum, stamps the precise transmit time, and sends it back. One safeguard is fundamental: the server refuses to answer at all until it has actually locked to a real time source. Before the first GPS or NTP sync it stays silent rather than handing out unverified, free-running RTC time that a client would trust. The moment a source locks, serving begins automatically.

Putting it together

On a typical boot the sequence is: the DS3231 provides a rough starting time from its backup battery (but the server does not serve yet); WiFi connects and, if configured, an NTP sample disciplines the clock within seconds; GPS then acquires a fix and takes over as the stratum-1 source, with NTP demoted to witness. From then on the served clock is anchored to the GPS 1PPS edge every second, holds over through brief GPS dropouts, and is continuously cross-checked against an independent Internet source — all while answering client requests on the LAN.

3. Hardware

The server is a single board built around an ESP32-S3 module. The major components and their roles:

Component	Role
ESP32-S3 controller	Runs the firmware: WiFi, NTP server, GPS parsing, info services, web interface, LED control
DS3231 RTC	Temperature-compensated real-time clock with coin-cell backup; the disciplined time source
u-blox GPS receiver	Supplies NMEA time messages and the 1PPS one-second pulse
BME280 sensor (optional)	Measures temperature, barometric pressure, and humidity; auto-detected at power-on and published on the network and status page. Omitted if not fitted
4 × APA102 LEDs	Two status indicators (top) and two color-configurable under-lights (bottom)
SW1 button	IP-address readout (double-click), AP mode (5-second hold), and over-the-air update (10-second hold)
USB-C	5 V power input

A coin cell on the DS3231 preserves the time across power loss, so after a reboot the server starts from a close approximation and is back to full accuracy as soon as a source re-locks.

4. First-Time Setup (AP Mode)

A brand-new server, or one that has been factory reset, has no saved WiFi credentials and so starts in Access Point (AP) setup mode automatically. In this mode the two status LEDs alternate steadily.

Connecting

From a phone or computer, open WiFi settings and join the network named InfoServer (or whatever Access Point Name has been configured). No password is required. Then open a browser to 192.168.4.1.

Network settings

- SSID — choose your 2.4 GHz home network from the dropdown. The ESP32 does not support 5 GHz networks.
- SSID Override — optionally type a network name by hand (for a hidden network, or one not in range during setup).
- Network Password — the password for the selected network.
- Access Point Name — the name this server advertises, used for the AP network, as its mDNS name (<name>.local), and as the hostname it reports to your router — so it appears by name, not as a generic device, in the router's client list.

Static IP (optional)

By default the server takes an address from your router by DHCP. Because clients are pointed at this server by address, a stable address is helpful; the most reliable approach is a DHCP reservation in your router. Alternatively, enable Use Static IP and fill in the IP address, gateway, subnet mask, and DNS server — choose an address outside your router's DHCP pool.

Finishing

Tap Save to store settings without rebooting, or Start to commit and reboot onto your network. After Start, reconnect your phone or computer to your normal WiFi; the server will appear at its DHCP or static address and as infoserver.local. (Once the server is running on your network, Save persists changes live and reboots automatically only when you change a setting that takes effect at boot — see Section 6.)

5. The Status Page

Browsing to the server's address shows a live status page that refreshes once per second without reloading. It is the primary window into what the server is doing.

Field	Meaning
UTC	The current served time in Coordinated Universal Time
Source	The active discipline source: GPS, NTP, or — (none yet)
Status	LOCKED (serving), HOLDOVER (GPS lost but riding the RTC), or NO SYNC
Last sync	How long ago the active source last corrected the clock
Clock temp	The RTC chip temperature, in the unit chosen on the Information page
Sensor	On-board sensor temperature, pressure, and humidity, in the units chosen on the Information page — always shown when a sensor is fitted, whether or not it is served at /info/sensor. An arrow beside the pressure shows the 3-hour trend (double arrow = fast); no arrow means steady
WiFi	Connected network and signal strength, or AP mode
NTP	The server's own IP address on your network
mDNS	The <name>.local address for discovery
Upstream	The upstream NTP server's hostname and reachability — OK, NO REPLY, or NO DNS — shown when an upstream server is in use
KoD	The last Kiss-o-Death code from the upstream server, with a count and age — shown only if one has been received
Served	Lifetime count of NTP requests answered, and the last client address
Uptime	Time since the server last achieved lock
Heap	Free memory and fragmentation (diagnostic)
Last reset	Why the server last restarted: power-on, software, panic, etc. (diagnostic)
GPS sats	In GPS mode: how many satellites are being used for the position fix, how many are being tracked, and how many are in view. At least four must be in use before a fix is possible
GPS signal	In GPS mode: the average and best satellite signal strength, in dB-Hz. Green is 35 or above, amber 25 to 34, red below 25. The figures rise as the receiver settles into a steady lock, so low readings shortly after

	start-up are normal
GPS data	In GPS mode: running counts of messages from the receiver — good, corrupted, with a fix, and without a fix (diagnostic)
GPS restarts	In GPS mode: how many times the receiver has reported starting up since the server was switched on. A small number at boot is normal; a count that keeps climbing means the receiver is restarting and will never hold a fix (diagnostic)
GPS jam filter	In GPS mode: whether the receiver accepted the interference-cancelling setting the server sends at start-up — enabled, rejected, or no reply. Should read enabled (diagnostic)
GPS-NTP	In GPS mode: the difference between GPS and the upstream NTP server, in milliseconds, while both are live (a dash otherwise). Turns red on a persistent disagreement over one second.

When a fault is active, a red warning line appears at the bottom of the box — GPS lock lost, holdover expired, upstream NTP unreachable, low memory, or Internet unreachable — and disappears the moment the condition clears. These are the same conditions that can notify you by email or Telegram (Section 8), shown here the instant they occur regardless of whether notifications are enabled.

6. Settings Reference

The web interface is organized into pages, reached from the buttons on the home (status) page: Settings (network and time), Information, More Sources (RSS feeds and nearby aircraft), Notifications, and — once the server is on your network — System. This section covers the Settings page; Information and More Sources are described in Section 7 and Notifications in Section 8.

These options appear on the setup form and, where they apply at runtime, on the live settings page. Save persists them. Most settings apply immediately and Save simply stores them; Save reboots automatically only when you change a setting that is applied at boot — WiFi network or password, the Access Point Name, any static-IP field, Discipline Mode, the NTP Server / Alternate NTP Server / NTP Polling Interval, or GPS baud. All other settings (holdover, LED brightness and color, everything on the Information page, and everything on the Notifications page) take effect without a reboot. When Save does reboot, the page waits for the server to come back and then returns you to the home status page automatically. If you changed the static IP, the page follows the server to its new address on its own, since that address is known. The one case it cannot follow is a change to your WiFi network or password: the server rejoins and is given a new address by your router that is not known in advance, so reconnect to it yourself at infoserver.local (or its new address).

Time source

Setting	What it does
Discipline Mode	NMEA GPS (default; GPS primary, NTP fallback) or Upstream NTP (no GPS; NTP is the only source)
NTP Server	Primary upstream time server, used as the fallback in GPS mode or the source in NTP mode
Alternate NTP Server	A second server the firmware rotates to if the primary is unreachable
NTP Polling Interval	How often to query the upstream NTP server (5 minutes to 4 hours)
NTP Fallback	How long GPS keeps discipline through a dropout before NTP takes over (1 hour to 1 day; default 4 hours)
GPS Baud	Serial speed of the GPS module (this receiver ships at 9600)

LEDs

Setting	What it does
Indicator LEDs	Brightness of the two top status LEDs: Disabled, Low, Medium, or High
Under-light Color	The color of the two bottom LEDs. The color also sets their brightness — a darker shade dims them, black turns them off

System

The System sub-page (available once the server is on your network) hosts the Firmware URL, the Update Firmware button, and Switch to AP Mode.

7. Information Services

Beyond serving time, the InfoServer can publish read-only information about its environment and location on your local network as simple JSON — so other devices, scripts, and dashboards can read it without any cloud service. This is the "info" in InfoServer. The available information includes readings from the optional on-board environmental sensor, current weather from Internet providers you configure, active National Weather Service alerts, computed sun and moon data, news headlines from RSS feeds you choose, and aircraft currently flying near you. Most sources are configured on the Information page of the web interface, which starts with the shared Location fields (latitude and longitude in decimal degrees) used by every location-based source; the RSS feeds and aircraft sources have their own More Sources page.

The /info manifest

Any device on your network can request <http://infoserver.local/info> (or the server's IP address followed by /info) and receive a manifest: a small JSON document listing the device, its firmware version, and every service currently available. The time service is always listed; each optional information source appears only while it has fresh data to offer. A client reads the manifest, then fetches whichever sources it cares about from the endpoints named there.

Endpoint	Provides	Appears when
/info	The manifest: device name, firmware version, and the list of services	Always
/info/sensor	On-board BME280 temperature, barometric pressure, humidity, and the 3-hour pressure trend	Enable Sensor is on, and a sensor is fitted and reading
/info/om	Current weather, today's and tomorrow's high/low and precipitation chance, and UV index from OpenMeteo	Enabled, with a location set and fresh data — no API key needed
/info/owm	Current weather from OpenWeatherMap	Enabled, with a key and a location set and fresh data
/info/vc	Current weather from Visual Crossing	Enabled, with a key and a location set and fresh data
/info/nws	Active National Weather Service alerts for the location	Enabled, with a location set and a fresh last check
/info/sun	Today's sunrise, sunset, day length, and the current moon phase	Enabled, with a location set and the clock synced
/info/rss	News headlines from up to	Enabled, with at least one

	five RSS or Atom feeds you choose	feed configured and fetching
/info/air	Aircraft currently flying within a chosen radius of the location	Enabled, with a location set and fresh data

Each reading is returned with an age in seconds, so a client can tell how current it is. A source with no data — no sensor fitted, no API key, no location, or data gone stale — is simply omitted from the manifest rather than returning an error, so clients degrade cleanly.

The environmental sensor

If an on-board BME280 sensor is fitted, the server reads temperature, barometric pressure, and relative humidity once a minute. The readings always appear on the status page — the sensor is detected automatically at power-on and simply omitted if absent. Serving them on your network at /info/sensor is controlled by the Enable Sensor toggle under Sensor on the Information page (off by default, like every information source): with it off, the sensor stays visible on the status page but is not published at /info. Each served value is offered in several units (for example temperature in Celsius, Fahrenheit, and Kelvin; pressure in pascals, hectopascals, and inches of mercury) so a client can use whichever it prefers.

The sensor also provides a classic barometer feature: the three-hour pressure trend. The server banks a pressure sample every ten minutes and serves the change over that window — the raw delta, the minutes it spans, and a plain word: rising fast, rising, steady, falling, or falling fast (a rapidly falling barometer is the time-honored sign that weather is about to turn). The status page shows the same tendency as a compact arrow beside the pressure reading — one arrow for a steady change, a double arrow for a fast one, none when steady. The trend needs history to be meaningful, so it reads "unknown" for the first hour after a restart and represents the full three-hour window after three hours.

Which units the status page itself displays is set under Sensor on the Information page: Temperature Units (Fahrenheit or Celsius, applied to both the Clock temp and Sensor readings) and Pressure Units (inches of mercury or hectopascals). This choice affects the status page only — the /info/sensor data always includes every unit regardless.

Weather sources

The server can fetch current conditions from up to three free Internet weather providers — OpenMeteo, OpenWeatherMap, and Visual Crossing — and republish them on your LAN at /info/om, /info/owm, and /info/vc. To enable a source, open the Information page in the web interface, enter your location, turn on that provider's Enable toggle, and — for

OpenWeatherMap and Visual Crossing — paste in the provider's free API key.
OpenMeteo needs no key at all, which makes it the quickest one to switch on.

- Location — your latitude and longitude in decimal degrees, at the top of the page and shared by every Internet source. In Google Maps, right-click your spot and click the coordinates to copy them.
- Enable toggle — each provider has its own on/off switch (off by default). Turning a source off pauses its fetching and serving without erasing the key below it, so it can be switched back on at any time.
- OpenMeteo — free worldwide service, no API key or account. Its data goes beyond current conditions: `/info/om` also carries today's and tomorrow's forecast high and low, the chance of precipitation for both days, and today's maximum UV index. (Precipitation probability is a forecast product the model occasionally omits; it is then served as -1, meaning "not provided.")
- OpenWeatherMap — paste a free key from openweathermap.org.
- Visual Crossing — paste a free key from visualcrossing.com.
- Poll interval — how often, in minutes, the server refreshes each provider (5 to 60). The defaults sit comfortably inside each provider's free tier.

A source that is toggled off, is missing its key, or has no location set is absent from the manifest. If a provider's data goes stale — older than an hour, for instance after an Internet outage — that source drops out of the manifest until a fresh reading arrives, so clients never receive outdated weather. Weather is fetched in the background and never delays time service.

NWS weather alerts

The server can also watch for official National Weather Service alerts — warnings, watches, and advisories — covering your location, and republish them on your LAN at `/info/nws`. The NWS service is free, needs no API key, and covers the United States and its territories. Turn it on with the Enable Alerts toggle under NWS Weather Alerts on the Information page; the Poll interval slider (2 to 15 minutes, default 5) sets how often the server checks.

Alerts are matched to your exact coordinates: the NWS resolves the Location fields at the top of the page to your county and forecast zone, and only alerts whose area actually covers that point are returned — a storm warning three counties away never appears. When more than one alert is active at once, the served data reports the total count and the details of the most serious one (ranked by severity, then urgency, then which ends soonest).

The `/info/nws` reading includes the number of active alerts, the top alert's event name (for example "Tornado Warning"), its severity both as a word (Minor, Moderate, Severe, Extreme) and as a number from 1 to 4 that simple displays can show directly, its urgency, certainty, and recommended response, the minutes remaining until it ends, and

its one-line headline. A count of zero is a confirmed all-clear — the source stays in the manifest with fresh data. Because a stale warning is worse than none, alert data older than 15 minutes drops the source from the manifest until the next successful check; with the poll interval capped at 15 minutes, a working connection never expires between checks.

Sun and moon

With the Enable Sun & Moon toggle on, the server publishes today's sunrise and sunset, the day length in minutes, and the current moon phase at `/info/sun`. Unlike the weather sources, nothing is fetched from the Internet: everything is computed on the server itself from your Location and the disciplined clock, using the standard astronomical formulas. That means it works with no Internet connection, has no key, no poll interval, and can never go stale — every request is calculated fresh at that moment, and the age reported is always zero.

Sunrise and sunset are served both as Unix timestamps (which a client device converts to its own local time) and as UTC clock strings. The moon is described three ways: its age in days through the 29.5-day cycle, its illuminated percentage, and its phase name — New Moon, Waxing Crescent, First Quarter, Waxing Gibbous, Full Moon, Waning Gibbous, Last Quarter, or Waning Crescent. At extreme polar latitudes, on days when the sun never rises or never sets, the day length is reported as -1 and the rise and set fields are blanked; the moon data is unaffected. The only requirement is a synced clock, so the source appears in the manifest shortly after boot once a time source has locked.

News headlines (RSS)

The server can follow up to five RSS or Atom feeds of your choosing and republish their latest headlines on your LAN at `/info/rss`. Most news sites publish a feed — look for an RSS link in the site footer, or search the site name plus "RSS". Configuration is on the More Sources page: turn on Enable RSS, paste a feed address into any of the five Feed URL slots (a blank slot is simply unused), and set Headlines per feed (1 to 8) and the Poll interval (15 to 120 minutes; feeds are cached by their publishers, so 30 minutes is a sensible default). Each feed is identified in the served data by the name the feed gives itself — there is nothing to type beyond the address.

The Test Feeds button fetches each configured feed once and reports the result slot by slot — the feed's name and how many headlines it offered, or the reason it failed (an HTTP error, an unreachable server, or a page that is not actually a feed). Nothing is saved by the test; it simply verifies the addresses as entered. Save also checks each address for obvious problems (a missing `http://` or `https://`, a stray space) and names the offending slot rather than storing a bad address.

The served data groups headlines by feed. Each feed appears with its slot number (its position on the form, 1 to 5), its name, and its most recent headlines as plain text —

cleaned of markup and typographic characters so simple displays can show them directly. A feed that is blank, failing, or stale — nothing fetched for over an hour — simply drops out of the served list without affecting the others, and clients address a feed by its slot number, which stays stable however many feeds happen to be serving. If every feed is out, /info/rss leaves the manifest entirely until one recovers.

Nearby aircraft

The server can watch the sky around you: it fetches live aircraft positions from a community flight-tracking network and republishes the aircraft currently within a chosen radius of your Location at /info/air. The data comes from ADS-B, the position broadcast that nearly all aircraft transmit, collected by volunteer ground receivers worldwide and aggregated by free community services. No API key or account is needed.

Configuration is at the bottom of the More Sources page. Enable Aircraft turns the source on; the Provider dropdown selects between two community networks, adsb.lol (the default) and airplanes.live — they publish identical data, so if one is ever unavailable, switching to the other restores the source immediately. Radius (5 to 100 miles, default 25) sets how far out to look. Minimum Altitude (0 to 10,000 feet in 500-foot steps) hides aircraft below that height — useful near an airport, where taxiing and pattern traffic would otherwise crowd the list; at 0 everything is included. Aircraft Shown (1 to 20, default 5) sets how many of the nearest aircraft are served, and the Poll interval (1 to 10 minutes, default 2) sets how often the picture refreshes.

The served data reports the total number of aircraft inside the radius, then details the nearest ones, closest first: callsign (for example AAL2279 — some private aircraft transmit none), aircraft type code, registration, distance and compass bearing from your location, altitude, ground speed, direction of travel, and whether it is climbing, descending, or level. Distances, altitude, and speed are each served in several units (miles, kilometers, and nautical miles; feet and meters; knots, mph, and km/h) so a client can use whichever it prefers. Aircraft move quickly, so the freshness rule is strict: data older than ten minutes drops the source from the manifest until the next successful update — with the poll interval capped at ten minutes, a working connection never expires between checks.

8. Notifications

The InfoServer can notify you when something affects its ability to keep good time — for example if GPS reception is lost or the upstream Internet time server becomes unreachable. Two delivery channels are available, and you can use either or both: email, sent through your mail provider, and Telegram, sent as instant messages from your own free Telegram bot. Both channels share one list of alert events — you choose what to be told about once, and every enabled channel delivers it. Notifications are off by default; enable and configure them on the Notifications page of the web interface.

How alerts behave

Alerts are designed to inform without nagging:

- One message per event. A standing problem — say, GPS staying lost for hours — produces a single message when it begins, not a repeat every few minutes.
- A 60-second settling time. A problem must persist for a full minute before it notifies, and must stay resolved for a full minute before the server considers it cleared. Brief momentary glitches are ignored.
- Grouped messages. If several problems occur close together (within about ten seconds), they are combined into a single message rather than arriving separately.
- Optional recovery notice. Turn on Notify on Recovery to also receive one message when a problem clears. Most faults send one; the one-time events — KoD Received, Device Rebooted, and Time Source Changed — have nothing to clear, so they never do.

Whether or not any notification channel is enabled, any active problem is also shown as a red line at the bottom of the status page the moment it occurs, and disappears when it clears (Section 5).

Setting up email (SMTP)

The server sends mail through a standard outgoing (SMTP) mail server — typically your email provider's. Fill in the SMTP Connection and Envelope fields:

Field	What to enter
Enable Email Alerts	Turn on to allow the server to send mail. When off, nothing is ever sent
SMTP Host	Your provider's outgoing mail server, e.g. smtp.gmail.com for Gmail
Security	The connection type, which also sets the port: SSL (465), STARTTLS (587), or None (25).

	Gmail uses SSL (465)
SMTP Username	Your login for the mail server — usually your full email address
SMTP Password	An app-specific password for the account (see below), not your normal login password
From Address	The sender address. For Gmail this must match the username
From Name	An optional display name for the sender (for example InfoServer)
Send To	Where alerts go. Separate multiple recipients with commas

Getting a Gmail App Password

Gmail and Google Workspace will not accept your normal account password from a device. Instead you create a one-off, revocable App Password. This requires 2-Step Verification to be turned on for the account first.

- 1. Turn on 2-Step Verification at myaccount.google.com → Security, if it is not already on.
- 2. Go directly to myaccount.google.com/apppasswords. Google no longer lists this page in the Security menu, so the direct address is the reliable way to reach it.
- 3. If prompted, re-enter your Google password.
- 4. Under "App name," type a memorable label such as InfoServer, and click Create.
- 5. Google shows a 16-character password in four groups of four. Enter it into the SMTP Password field without the spaces.

Use your normal Gmail address for both the SMTP Username and the From Address. If the App Passwords page says the option is not available, 2-Step Verification is not fully enabled yet, or an administrator has disabled app passwords on a Workspace account.

Setting up Telegram

Telegram is a free messaging app (telegram.org) for phones, tablets, and computers. As an alert channel it has some real advantages over email: messages arrive instantly as notifications on every device you run Telegram on, nothing lands in a spam folder, and there is no mail server or password to configure. Alerts are sent to you by your own Telegram bot — a private automated account that you create once, in about two minutes, using the steps in the next section. The server needs just two pieces of information, both on the Notifications page:

Field	What to enter
Enable Telegram Alerts	Turn on to allow the server to send Telegram messages. When off, nothing is ever sent
Bot Token	The secret token BotFather issues when you create your bot — the bot's password. See the steps below

Chat ID	The numeric Telegram ID of the person (or group) that receives the alerts. Not your phone number — press Get Chat ID and the server fills this in for you
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Two buttons under the fields complete the setup: Get Chat ID looks up and fills in the Chat ID automatically, and Send Test Telegram confirms the whole channel end to end.

Getting the Bot Token and Chat ID, step by step

Bots are created by talking to BotFather — Telegram's own official bot-creation assistant. The whole process happens inside the Telegram app; nothing is installed and nothing is paid for.

- 1. If you do not already use Telegram, install it from telegram.org (or your device's app store) and sign in with your phone number.
- 2. In Telegram's search box, type BotFather and open the result named BotFather with the blue verified check mark (its full handle is @BotFather). Tap or click Start.
- 3. Send it the message /newbot. BotFather first asks for a display name for your bot — this is just a label, so type anything you like, for example InfoServer Alerts.
- 4. It then asks for a username for the bot. This one must be unique across all of Telegram and must end in the letters bot — for example SmithInfoServer_bot. If your first choice is taken, try a more distinctive one.
- 5. BotFather replies "Done!" with a long access token that looks like 8123456789:AAHrX9k2mQ... — this token is the Bot Token. Tap it in the chat to copy it.
- 6. Open the InfoServer's Notifications page and paste the token into the Bot Token field.
- 7. Back in Telegram, open a chat with your new bot — BotFather's "Done!" message includes a t.me link to it; tap that link, then tap Start (or simply send the bot any message, such as "hello"). Do not skip this: the server can only discover your Chat ID after the bot has received at least one message from you.
- 8. On the Notifications page, press Get Chat ID. The server asks Telegram who last messaged the bot and fills in the Chat ID field, confirming with your name — for example "Chat ID 123456789 (Sam) — saved." The ID is stored immediately; you do not need to press Save for it.
- 9. Turn on Enable Telegram Alerts and press Send Test Telegram. The test message should appear in your Telegram chat within a few seconds.

Keep the token private — anyone who has it can send messages as your bot. If it ever leaks, send /revoke to BotFather to invalidate it and issue a fresh one, then paste the new token into the Bot Token field.

To deliver alerts to a family or team group instead of one person, add the bot to the group, send any message in the group, and press Get Chat ID — group IDs are negative

numbers, which is normal. And if Get Chat ID ever reports that no messages were found, simply send the bot (or group) another message and press the button again: Telegram only holds a bot's undelivered updates for about 24 hours.

Choosing which events notify you

Each event type has its own on/off toggle; all are on by default. Turn off any you do not want to hear about. The event list is shared by both channels — each event you leave on is delivered by email, Telegram, or both, according to which channels are enabled.

Event	Alerts when
GPS Lock Lost	GPS drops a valid fix and the server falls back to the DS3231 holdover
Holdover Expired	The holdover window runs out and the server switches to the Internet NTP fallback
NTP Unreachable	The upstream NTP server cannot be resolved or stops replying
Internet Down	WiFi is up but nothing beyond your router responds. The server probes the Internet every 5 minutes and the fault trips after two consecutive misses. The outage itself usually blocks the onset message — with Notify on Recovery on, the "Internet restored" message is the one that reliably arrives
KoD Received	The upstream NTP server sends a Kiss-o-Death asking the server to back off; the message includes the code, with a plain-English explanation for the common ones (RATE, DENY, RSTR)
Device Rebooted	The server restarts — one message, including the reason for the restart
Low Heap / Health	Free memory falls to a level that may affect stability
Time Source Changed	The served time source flips between GPS and NTP in either direction — GPS to NTP when GPS is lost past the holdover window, and NTP back to GPS when GPS returns. One message per switch
Weather Alert	The NWS issues a new alert for your location at or above the Minimum Severity set below it. Requires NWS Weather Alerts to be enabled on the Information page. One message per new alert, with its headline and severity

The Weather Alert event has one extra setting, Minimum Severity: Any, Moderate, Severe, or Extreme. Only alerts at or above the chosen level are sent — set it to Severe, for instance, to hear about severe thunderstorm and tornado warnings but not routine advisories. The choice affects notifications only; the `/info/nws` data always carries every active alert regardless. With Notify on Recovery on, one further message arrives when the last qualifying alert has cleared.

About time-source switching

In NMEA GPS mode the server normally disciplines its clock from GPS. When GPS reception is lost it does not switch away immediately: it rides the battery-backed DS3231 holdover — still GPS-anchored and still stratum 1 — for the configured NTP Fallback window (Section 6). Only if GPS has not returned by the end of that window does the served time source actually switch to the upstream Internet NTP server. When GPS later reacquires a fix, the source switches back to GPS. Each of these switches is a change in which source your devices are being served from.

The alert events cover this in two complementary ways, so you can pick the level of detail you want:

- Time Source Changed fires once on every actual switch, in both directions — a single "GPS to NTP" or "NTP to GPS" message. This is the simplest way to be told only when the source your clients rely on has really changed.
- GPS Lock Lost and Holdover Expired break the GPS-to-NTP direction into its two stages: GPS Lock Lost the moment the fix drops (the server is now on holdover), and Holdover Expired when the window runs out and NTP actually takes over. The return to GPS is reported by Time Source Changed, and — if Notify on Recovery is on — also as "GPS lock restored" / "Holdover recovered".

Because these overlap, a single GPS-to-NTP switch can, with everything enabled, produce a Time Source Changed message and (about a minute later, after its settling time) a Holdover Expired message. If you want only one, leave Time Source Changed on and turn Holdover Expired off, or vice versa. In pure Upstream NTP mode there is no GPS, so no source switching occurs and none of these fire.

Testing

Each channel has its own test button. After filling in the email connection and envelope fields, use Send Test Email; after setting up Telegram, use Send Test Telegram. In both cases the server sends a test message and reports the result on the page — either confirmation that it was sent, or the specific error that came back: a wrong password or host for email, an invalid token ("Unauthorized") or wrong chat ID ("chat not found") for Telegram. This is the quickest way to confirm your settings before relying on alerts. The tests work even while a channel's master Enable is off, as long as its fields are filled in.

9. Indicator LEDs and Under-light

The server has four LEDs in one chain: two status indicators on top and two under-lights on the bottom.

Status LEDs (top)

The two top LEDs report what the server is doing. The STATUS LED pulses one short heartbeat per second; its color tells you the active state — green for GPS, blue for NTP, amber for GPS holdover. The ACTIVITY LED flashes that same color as each time request is served, and flashes purple when an information request is served.

State	STATUS LED	ACTIVITY LED
Starting up	● Red — solid until the first time source locks	Off
GPS locked	● Green — one short heartbeat pulse per second	● Green flash on each NTP time request served
NTP locked	● Blue — one short heartbeat pulse per second	● Blue flash on each NTP time request served
GPS holdover	● Amber — one heartbeat pulse per second	● Amber flash on each NTP time request served
Information served	Heartbeat continues (source color)	● Purple flash on each /info request served
No sync	Off	Off
AP setup mode	● Red — alternating, ½ second	● Red — alternating, ½ second
Firmware update	● Solid red	● Solid red
IP-address readout	Blinks the IP out octet by octet (Section 10)	Alternates green/red with STATUS to mark each digit

At power-on the STATUS LED is solid red until the server locks to its first time source, at which point it switches to the green, blue, or amber heartbeat above. The ACTIVITY LED gives a brief flash on each request answered: an NTP time request flashes in the same color the STATUS LED is showing (green on GPS, blue on NTP, amber in holdover), while an information-service request (the /info endpoints, Section 7) flashes purple — so you can tell time traffic from info traffic at a glance. The Indicator LEDs setting (Section 6) dims or disables the normal GPS, NTP, holdover, and activity indications; the attention states — AP setup mode, firmware update, and IP-address readout — always show at full brightness regardless of that setting.

Under-light (bottom)

The two bottom LEDs are decorative under-lighting set to a single constant color by the Under-light Color picker (blue by default). Because the color carries its own brightness, picking a darker shade dims the under-light and choosing black turns it off entirely. The under-light is independent of the Indicator LEDs brightness setting.

10. The SW1 Button

Action	Result
Double-click	Blink out the server's IP address on the LEDs, one octet at a time
Hold 5 seconds	Restart into AP (setup) mode temporarily, to reconfigure WiFi or settings — the full setup form appears, and Start (or a power cycle) returns the server to normal operation
Hold 10 seconds	Check for and install a software update over WiFi (see Section 11)
Hold during power-on	Factory reset — wipe all settings and reboot into AP setup mode

The two timed holds share one button, so the duration matters. Release at about five seconds for AP mode; keep holding to about ten seconds for the update check. Watch the LEDs: both go solid red when the update begins. A hold during power-on (before the LEDs come up) is the factory reset.

Reading the IP blink

After a double-click, the IP address is signalled octet by octet. Each digit is shown as that many blinks (a zero is shown as a single long flash), with the two LEDs alternating to mark digit boundaries, and a longer pause between octets. For example, 192.168.4.75 blinks as 1-9-2, then 1-6-8, then 4, then 7-5. For most users the status page is an easier way to read the address; the blink readout is there for when no screen is handy.

11. Over the Air Updates

New firmware can be installed over WiFi without a cable. Hold SW1 for ten seconds while the server is running. (A shorter, five-second hold switches to AP mode instead — see Section 10, so keep holding past five seconds for an update.) Both status LEDs go solid red while the update runs.

The server fetches a small version file from the configured Firmware URL and compares it with the running version. If they match, it reports that it is already current and resumes normal operation. If a newer version is available, it downloads and installs it, then reboots into the new firmware. The whole process is automatic once triggered; do not remove power while both LEDs are solid red.

The Firmware URL is set on the System sub-page and points at the location that hosts the version file and firmware image. The Update Firmware button on that page does the same thing as the 10-second button hold, from the comfort of the web interface.

12. Specifications

Item	Specification
Controller	ESP32-S3 (240 MHz dual-core, WiFi)
Time source	DS3231 TCXO RTC (± 2 ppm), battery-backed
Reference (GPS mode)	u-blox GPS, NMEA + 1PPS — stratum 1
Reference (NTP mode)	Upstream NTP server — stratum 3
LAN accuracy (GPS)	Approximately ± 1 ms served; clients typically see ± 15 ms end to end over WiFi
Holdover drift	About 7 ms per hour on the DS3231 after GPS loss
Protocol	NTP v4 (RFC 5905) over UDP port 123
Environmental sensor	Optional BME280 — temperature, barometric pressure, and humidity
Info services	JSON /info manifest publishing on-board sensor readings with pressure trend, OpenMeteo / OpenWeatherMap / Visual Crossing weather, NWS weather alerts, computed sun & moon data, RSS news headlines (up to 5 feeds), and nearby ADS-B aircraft
Notifications	Email (SMTP — SSL, STARTTLS, or unencrypted) and Telegram bot messages on time-source faults
Network	2.4 GHz WiFi; DHCP or static IP; mDNS discovery
Power	5 V USB-C

13. Troubleshooting

Symptom	What to check
Source stays at — / NO SYNC	In GPS mode, give the unit a clear view of the sky and allow several minutes for a first fix. Confirm WiFi is connected so the NTP fallback can also work.
Never gets a GPS fix	Move the unit to a window or outdoors; cold starts under cover can take a long time or never complete. The status page Source will read NTP while GPS is unacquired. Check GPS sats — four must be in use before a fix is possible — and GPS signal, which should climb well above 25 dB-Hz with a clear view of the sky.
GPS restarts keeps climbing	The receiver is restarting instead of running continuously, and cannot stay locked long enough to complete a first fix. The count should settle within a few seconds of switching on and then stop. If it keeps rising, make sure the unit is running the current firmware.
GPS signal stays low	Weak readings are normal indoors and immediately after start-up, and they climb as the receiver settles into a continuous lock. Persistently low figures alongside a climbing GPS restarts count point to the receiver restarting rather than to poor reception.
Status shows HOLDOVER	GPS lost its fix but the server is still serving GPS-derived time from the RTC. If it persists, improve the unit's view of the sky; after the NTP Fallback window the server switches to NTP.
GPS-NTP shown in red	GPS and the upstream NTP server disagree by over a second. Suspect a GPS fault, poor reception, or a bad upstream server; try an alternate NTP server.
Clients can't reach the server	Confirm the server's IP on the status page and that clients point at it. A DHCP reservation or static IP keeps the address stable.
Can't find the server on the network	Try http://infoserver.local/ (or the configured name). If WiFi setup failed, the unit falls back to AP mode — rejoin the InfoServer network and reconfigure.
A weather source is missing from /info	Confirm that provider's API key and your latitude/longitude are set on the Information page, and that the server has Internet access. Data over an hour old is dropped until it

	refreshes.
/info/nws is missing from the manifest	Confirm NWS Weather Alerts is enabled and a latitude/longitude is set on the Information page, and that the server has Internet access. The NWS covers US locations only, and alert data over 15 minutes old is dropped until the next successful check.
A feed is missing from /info/rss	Press Test Feeds on the More Sources page — it reports each slot's result. Confirm the address is the feed itself (not the site's home page) and that the server has Internet access. A feed that has not fetched for over an hour is dropped until it recovers.
/info/air is missing from the manifest	Confirm Enable Aircraft is on and a latitude/longitude is set on the Information page, and that the server has Internet access. If the selected provider is down, switch Provider on the More Sources page to the other network. In quiet airspace also check the Minimum Altitude filter — set high, it can leave nothing to serve while the count stays honest.
Internet unreachable shown in red	WiFi is working but nothing beyond your router responds — usually the modem or the provider. Check the modem and the router's Internet (WAN) light. The warning clears itself a few minutes after the connection returns; with Notify on Recovery on, an "Internet restored" message confirms it. The probe makes a TCP connection to a public DNS server (port 53), so a firewall blocking outbound DNS can cause a false alarm.
Test email fails	Check the SMTP host, Security/port, username, and password. For Gmail use a 16-character App Password (Section 8), not your account password, and make sure From matches the username.
No alerts arrive	Confirm the channel's Enable toggle (email or Telegram) and the relevant event toggle are on, and that its test button succeeds. Remember a problem must persist about a minute before it notifies.
Test Telegram fails	"Unauthorized" means the Bot Token is wrong or was revoked — re-copy it from BotFather. "Chat not found" means the Chat ID is wrong — run Get Chat ID again (Section 8).
Get Chat ID finds no messages	The bot has not received a message recently — Telegram holds a bot's updates for only about a day. Send your bot any message in Telegram, then press Get Chat ID again.
Both LEDs alternate fast and won't stop	The unit is not provisioned (authenticity record missing). It must be provisioned before the firmware will run.
Forgotten settings / start over	Hold SW1 while applying power to factory reset into AP mode.

Notices

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Power the server only from a standard 5 V DC USB-C source. The InfoServer contains no user-serviceable parts.

InfoServer — Designed by Mitchell Feig